Abstraction removes unnecessary detail to make complex process easy to understand

Applications *Brookshear, J.G. and D. Brylow Computer science: an overview*

It contains code in an executable file loaded into memory together with related data like resource files and libraries and possibly another file that the user is working on

Resource files - Type of data file that is not part of the EXE and contains non-executable data an application needs, such as images, icons, menus, dialog box layouts, sounds, and localized strings

Dynamic libraries (also known as a shared library, .dll in Windows, .so in Linux, .dylib in macOS) - Collection of compiled executable code/functions and data that can be used by multiple programs simultaneously

API - Version of the website that is designed for computer programs and not humans. They are how we have different versions of the same application on different devices, mobile, web-based or across different OS. App Developers create APIs that can display data in different ways, in that application form or in a website.

Debugging- **Understand the system** and **look, don't think**, can only understand a problem by looking at what the computer is doing, then **make it fail**, you need to know what cause it to fail in the first place by reproducing the bug reliably and **only change one thing at a time**, which lets you gather more information on the issue

CPU and Memory

CPU – Most important component that does all the calculations and instructions that make up programs we call code. **Arithmetic Logical Unit** performs numerical mathematical calculations. **Control Unit** controls the running of programs

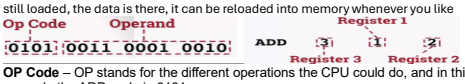
RAM– Stores all the data that is being worked on

Example: You're editing a document, it will be stored in RAM, if you're watching a video, it will pass through RAM before going to the screen

DMA - A technique allowing hardware components to read/write to memory directly, bypassing the CPU to improve system efficiency

Virtual memory – It's disc space that is acting as memory. The process of copying memory to hard disk is called paging, data will be slower to access but it frees up space on RAM

Example: Memory manager takes a running application that's not active now and moves the memory area of that application from main memory onto the hard disk. It's still loaded, the data is there, it can be reloaded into memory whenever you like



OP Code – OP stands for the different operations the CPU could do, and in this example the ADD code is 0101

Operand – Is data that is being used in the operation, which is stored as binary Bit Patterns, any instruction will depend on the data, so the operation adds number that is stored in Register 1 and add that to Register 2 and put the result in Register 3

Arithmetic Operations: | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |

ADD: Adds two operands together.

SUB: Subtracts one operand from another.

MUL: Multiplies two operands.

DIV: Divides one operand by another.

INC/DEC: Increment or decrement the value of an operand by one

Data Transfer Operations:

MOV: Moves data from one location to another.

LOAD: Loads data from memory into a register. Last two parts of the operand are combined to give you a memory address as there is a lot of memory addresses unlike Registers

MEMLOAD: Transfer data from a main memory location into a register for processing

VALLOAD: Load a direct number/value or a specific address value into a register

STORE: Stores data from a register into memory. (Reverse from LOAD)

PUSH/POP: Pushes data onto or pops data off a stack

Control Flow Operations:

JUMP (JMP): Changes the flow of execution to another memory location, jumping to a different instruction in memory, changing the program counter, instead of adding one it jumps using last two parts of the operand and copies them into the program counter

CONDITIONAL JUMP: Jumps to a different location based on a specified condition (e.g., jump if equal, jump if not equal). It takes the value in the register given by the first operand, in this case register 5, and compares it to the value of register zero. It's always register zero

CALL/RETURN: Calls or returns from a subroutine or function

Logical Operations:

AND/OR/XOR/NOT: Perform logical operations on bits or bytes.

SHIFT/ROTATE: Shifts or rotates bits left or right

Comparison Operations:

CMP: Compares two operands.

TEST: Performs a bitwise AND operation to test bits

Attributes	Registers	Cache (L1, L2, L3)	RAM	Mass Storage
Location	On CPU	On or Close to CPU	Standalone	Standalone
Capacity	Very small (KBs)	Small to mid (MBs)	Mid to Large (GBs)	Large (TBs)
Speed	Extremely fast	Very fast	Fast	Slower than others
Access time	Nanoseconds	Nanoseconds	Milliseconds	Seconds
Hierarchy level	Highest	Middle	Middle	Lowest
Purpose	Store data for immediate use	Store frequently used data	Bridge between CPU and Storage	Long-term storage of data
Volatility	Volatile	Volatile	Volatile	Data persist when off
Cost	High	Moderately high	Moderate	Low/Lower

(L1 being smallest in capacity and fastest, L3 is larger but slower)

Bus – Wiring that connects the CPU and Memory and allows fast interchange between the two and allows for connections to peripheral devices, there is a **controller** (Hardware/Software device) in between the device and the bus

USB – Type of powerful controller in the computer that allows data to be transferred between the CPU and devices independent of the details of the exact device

Memory Mapped IO - Where CPU peripheral devices (like graphics cards or keyboard) are mapped to specific main memory address, allowing them to be accessed using standard **LOAD** and **STORE** instructions or Both

Data Representation

Bit - Something that can only have one of two states (1 or 0) true or false, an abstract representation of any binary

Byte – Is 8 **Bits**, 256 different patterns of ones and zeros called bit patterns, each pattern can represent something different.

Example: Single bit can represent black or white, but putting together lots of black and white pixels, you can make a picture. The same bit pattern can also represent a letter G, the number 71 or dark Gray

ASCII - Limited character set for basic English (**Fixed-length, 128 characters, 7/8 Bits**)

Unicode - Universal, expansive **Encoding** standard, supports all writing systems and is backward-compatible with ASCII (**Over 149,000 Characters, Variable, 16/32 Bits**)

Brightness - Typically represented by eight bits, shades between black and white will be a number between 0 and 255. Colour, we use **Three** numbers, Red, Green and Blue, **Fourth** number, Alpha, is added to allow for transparent images. Alpha of 0 means a completely transparent pixel and 255 is completely opaque

Audio - Can record the air pressure change as a number. For realistic sound, must be done this very fast over 44,100 times a second **(44.1kHz).** One of these individual recordings is called a sample and a whole sequence of samples creates a sound file

Lossy – Some Data will be lost, used for Video and Audio as Humans are not likely to notice the lost details, **DCT**, **Vector Quantisation** and **Transform Coding**

Lossless – When Data Integrity is important and needs preserving like a Text File, you don't want words to start disappearing, **RLE**, **String-table compression**, **LZW** and **zlib**

	Example input	Encoded form	Why it is efficient	Best used for
RLE	11111100011111111111111111	001101000110100011	`00110` = 6, followed by 1; then `00011` = 3, followed by 6; then `10001` = 17, followed by 1.	Binary data or images with large, repeated areas
LZW	aabaaaba where `0` = a, `1` = b	`001231`	Here, the extended dictionary includes 2 = aa and 3 = ab, so aabaaaba can be stored more compactly than `00100010`.	More complex data such as gradients in images and text

Compilers - Take a high-level language, convert it directly into machine instructions

Interpreters – That instead of converting code all at once, will convert a language like JavaScript into machine instructions as it's running

JPEG	Lossy Compression	.jpg, .jpeg	Photographs, web images, digital media
PNG	Lossless Compression	.png	Graphics, Logos, Images with Transparent
GIF	Lossless or Lossy Compression	.gif	Animated Images, Simple Graphics
TIFF	Lossless or Lossy Compression	.tiff, .tif	High-quality prints, archival storage, professional use
BMP	No Compression	.bmp	Legacy applications, uncompressed storage
WEBP	Loss or Lossless Compression	.webp	Web images, photography, transparent images
JPEG 2000	Lossy or Lossless Compression	.jp2, .j2k, .jpf, .jpx	Medical Imaging, satellite imagery
HEIF	Lossy or Lossless Compression	.heic, .heif	High-efficiency format used in mobile devices

	MP3 lossy	AAC lossy	FLAC lossless	WAV lossless	RAW lossless	AAX lossy
quality	✓	✓	✓	✓	✓	✓

low weight	✓	✓			✓
compatibility	✓	✓	✓	✓	
metadata	✓	✓			✓

Operating System (OS)

It allows you to communicate with the computer without knowing the computer's language, it manages the computer resources, establish a user interface, executes and provide services for applications, it also protects data with security

Utilities – Little applications that are bundled with the **OS but not part of it**

Example: Calculator, calendar, to-do list and the ability to set the time

GUI – Menus and taskbar dock

Example: Window containing the browser is supplied by the Window manager

GUI Library – Where Buttons or text entry boxes are supplied from, some basic button functionalities are shared with application in the OS

Windows Manager - **Part of the OS** that provides space on screen

Files Manager – **Part of the OS** that manages the files and the **metadata** of the files on disk, it also manages **Folders** which are also called **Directories**

Memory Manager – **Part of the OS** that allocates memory to applications and making sure that one application can't interfere with the memory of another application, so it can stop one application from accessing the memory area of another application. It defines what memory areas are accessible to each app, and what area of access is allowed by that app

Kernel – Is the **Core of the OS** and have complete control over the system, it is the portion of the operating system code that is always in memory, the first program that is loaded into memory, the Kernel is also responsible for manging device **Drivers**

Drivers – Specific type of software that acts as a translator between operating systems and hardware peripherals | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |

System Metadata Example: Name, size, file path, creation/modified dates and attributes

Content Metadata – Specific to file type. **Example**: camera model, ISO of a photo or Word count and template name for a document

OS Processes

Process - Contains code that runs independently of other code and has its own memory

Treads – Runs independently at the same time as other bits of code but doesn't have its own memory, it **shares memory with other threads**

CPU Time Slice – Processes run one after another, only runs for a certain amount of time, after which it gets an **Interrupt** and move to the next process, all happens in **10-100 milliseconds**. If time slice is too short, scheduler will take more processing time. If the time slice is too long, scheduler will again take more processing time.

Interrupt - Message sent to the process that says stop, get off the CPU

Example: Loading off a disk is much slower than running things on the CPU. While it's waiting for that, the CPU **Interrupts** that process and moves on to different process. When the data arrives, **Interrupt**. The previous process finally got the data and can start again.

STCF Optimal Scheduling - Run the thread that will finish fast without interruptions. It can also give priority to I/O-bound jobs, which keeps the disks/network as busy as possible

Process	Thread
Instance of a program being executed	Segment of a process managed by scheduler
Each treated as a new process by the OS	OS takes all User-level threads as a single Process
One gets blocked by the OS, the other continues	One gets blocked by the OS all gets blocked
Context switching is longer as they are heavy	Context switching is fast as they are lightweight
Data and code Segment of each are independent	Data and code Segment of each are shared
The OS takes more time to terminate	The OS takes very little time to terminate
New Process creation takes time and all resources	New Thread creation takes less time

Semaphores - Special instruction in hardware that first checks whether a lock is locked and sets the lock at exactly the same time as it checks, so that nothing can interrupt that interaction, this instruction must be implemented directly in the CPU

Deadlock – Happens when the first process wants/waiting for the second resource and the second process wants/waiting for the first resource. It can't proceed and won't unlock its resource. So, both processes are waiting for the other one to unlock their resource.

Example: Process 1 might want to play the file on the sound card, but it's locked. Process 2 might want to record to that file, but it's also locked.

Coffman Conditions – Deadlock can occur when **Mutual Exclusion,** which states each resource is assigned to one process at a time, but devices like printers and tape drives, require exclusive access by a single process, and **Circular Wait,** means a process is holding and wants more of that resource that is being held by other processes, and **Hold and Wait**, means when one or more processes can hold and wait for other resources to become available for use, and lastly **No Pre-emption**, means resources that have been granted access in advance, cannot be taken away at that time, **all to occur simultaneously together**

Solving Deadlock: Deadlock detection and recovery - Abort a process or pre-empt some resources when deadlocks are detected

Ignoring the Problem all together – Just reboot the system

Avoid Circular Wait - Number all resources, and require that process to request resources only in strictly increasing or decreasing order

Avoid Hold and wait - Processes must be prevented from holding one or more resources while simultaneously waiting for one or more of others

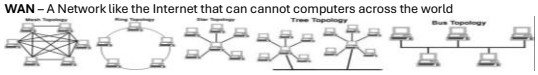
Avoid No Pre-emption - Process is forced to wait when requesting a new resource, and all other resources previously held by this process are released (Pre-empted), forcing this process to re-acquire the old resources along with the new resources in a single request

Networks

PAN – Simple short-range, low-power connections, like Bluetooth Networks between phone, laptop, smartwatch

MAN - High-speed network that connects multiple **LANs** a city or large campus

WAN – A Network like the Internet that can cannot computers across the world



Ring - Devices are connected in a closed circle, and data travels (clockwise or counter-clockwise), often using token passing.

Tree - Connects multiple star networks to a main bus cable with a core backbone switch (root) connecting distribution switches (branches), which then connect to individual workstations (leaf nodes)

Bus - All nodes connect to a single central cable, known as the backbone, with terminators at both ends, early local area networks, rarely used today.

Client-Server Communication – Individual Computers (Clients) all communicate with a central computer (Server)

Star Network - Clients and servers are not directly connected to each other. They are connected in complex ways over the Internet. So, it isn't about physical connections but about virtual connections using network protocols

Cluster - Computers are networked together and do their work in parallel, like a single combined computer with a lot of redundancies. Web servers are implemented as a cluster

Virtual Servers – Implemented using **Virtual Machines** or **Containers** which are more efficient, they are like Virtual Machines, but they share OS Kernels,

Peer-to-peer - Individual people's computers communicate directly with each other

Routers – Can understand many network protocols and link 2 networks together translating packets between different protocols usually in home networks

Switches – Can understand one protocol very well and uses Ethernet Cables to link devices together, typically forwarding packets unchanged, usually in Office Networks

ISP - The essential bridge connecting LAN to the global, public network (WAN). These networks can span whole countries and are evolution of the old-wired telephone networks

IP Address - Every computer on the Internet has one, which allows the router to know how to reach it

URL – More user-friendly type of address for humans to remember and search for, but more complex for routers

Domain Name - Is converted into IP address using Internet service call DNS

Carrier Sense Multiple Access with Collision Detection (For Ethernet) – When a computer transmits, it also checks for other transmissions on the Bus, and will stop both transmission with the same protocol and each computer will wait for a random period of time till they transmit again and since it is random it's unlikely that the other computer will transmit at the same time again

MEAN stack - A full JavaScript web stack, MongoDB (Non-Relational Database), Express (Server-side Web Framework), Angular.js (Front-end development/UI), and Node.js (Server Runtime for Web Servers). It is popular because one language, JavaScript, is used across client, server, and much of the data flow, which simplifies full-stack development.

LAMP stack - A traditional web stack, Linux OS, Apache (Web Server), MySQL/MariaDB (Relational Database Management System), and PHP (Server-side scripting language; sometimes Perl or Python is used for the "P"). It is widely used for database-backed websites because each layer has a clear role: OS, web server, database, and back-end logic.

Quick Contrast: MEAN is modern, JavaScript-based, and commonly used for dynamic single-page apps; LAMP is older, modular, and strongly associated with classic server-rendered websites.

Databases (DB) – Collection of data stored in a structured way, each row of the data table is a different entry called **Tuple**, each column of a entry is called **Attribute, SQL** is Table-based, Vertical scaling, **NoSQL** is non-relational, Horizontal scaling which is cheaper

Relational Database Command (Join) – Allows you to take data from one table and join it with data in another table to create a new combines table. Most relational databases use mainly a programming language called SQL as the way to access these commands

Security

Virus - Piece of software that copies itself from one computer to another without your knowledge, and aims at damaging your computer in some way, and can steal your data

Spyware - Records what you do and sends the information to a third party

Trojan Horse - Disguise to be legitimate software, perform harmful actions in secret

Phishing Attack - People are fooled into giving away important data like passwords or bank details (Scam Site)

Trap Door – Security Breach without your knowledge, and can harm the files and data

Worm – Destroys a system by using system resources to extreme levels and can deny other processes from access, it can shut down entire networks

DDoS – Denies the user access to the system and overwhelms the system with request, breaking the system

Anti-Virus Software – Updated with information about all known viruses and searches your computer for them

Firewall - Prevents certain types of packets from coming in/out of a machine or a network, which prevents hackers hacking your computer or **DDoS.** If configured correctly with proxy servers, it is good defence against a wide range of threats

Proxy Servers - A computer that acts as a link between a client and a server or between a client and computers on a network, all traffic must go through it first. Forward proxies sit in front of clients to manage outbound traffic. Reverse proxies sit in front of web servers to manage inbound requests.

Access Control - Only certain people can have access to a computer, a server or a website through a password and two-factor authentication, requires you to enter a code that the site sends to your mobile phone

Users – Have Limited to install software or make changes to the computer and can't access another user files

Administrators – Have complete rights to the computer and can install applications and add/delete user accounts, and sometimes they can change user files

Privileged Instructions – Instruction with elevated privileges that can only be executed by the OS, particularly the kernel in **Supervisor Mode** or **Privileged Mode**. This level of security is built into the **hardware of the CPU** and **not part of the OS**

Example: Application running on the OS needs to perform an operation that requires special permissions, like accessing hardware directly or modifying system settings, it invokes a system call to the kernel. The kernel, in **Privileged Mode,** performs the requested operation on behalf of the application

Privileged Code – **Part of the OS** that runs in high-privilege/Kernel Mode, acts as the supervisor, while **Privileged Instructions** are hardware-enforced, machine-level

Email Protocols - Email is not distributed by the Web, but can via webpage

SMTP – Standard protocol for sending email from a client to a server or between servers, for outgoing mail

IMAP – Receive and manage emails directly on the server, allowing synchronization across multiple devices

POP3 – Downloads emails from the server to a single local device, usually deleting the original from the server, for offline access

Web Protocols - Webpage for an organisation, data is transferred between a server (Computer that holds information that is used by many clients) and a client (Web Browser/User Computer)

VoIP/VoLTE – **Internet Telephony**, peer to peer phone conversations are sent using the Internet protocol

TCP – Guarantees delivery of message and is a reliable protocol with handshakes, used for financial transactions

UDP - Doesn't guarantee delivery of messages but provides instant update, used for gaming or video conferencing

HTTP and HTTPS - The S stands for secure, encrypted version where all data transmitted is encrypted, most browsers will show a little padlock sign. Involves a client sending a HTTP request for a web page to the server, which creates the page and sends it back

HTML - Foundational language for creating and structuring WWW content/webpages

XML – For **storing, transporting, and structuring data** in a format that is readable by both humans and machines, unlike HTML which uses fixed set of tags to structure content, XML allows users to define their own custom tags

JSON - Lightweight, text-based format used primarily for transmitting and storing structured data **alternative to XML** for APIs, configuration files, NoSQL databases

Cookies - Designed to keep track of the state of the websites, they are small amounts of data that websites store in our computers as we use those sites, they are stored by browser software locally, not on any server

Example: How websites track whether we're logged in or not, and how they know our browsing history

Network Protocols: **Application Layer** – Handles communication specific to the application over the web (**HTTP/S, SMTP, IMAP, POP3, DNS**)

Transport Layer – Handles end-to-end communication between the 2 computers that are exchanging messages (**TCP** and **UDP**)

Network Layer – Handles communication across networks via routers (**IP**)

Link Layer – Handles Physical networks via switches, bridges, and network cards (**Ethernet, Wi-Fi**) | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.1 |

Machine Learning (ML) – **1. Data Collection & Preparation**: Gather relevant data, removing duplicates, selecting important variables to improve model performance.

2. Model Training & Selection: Select algorithm and feed the Training Data Set **(4/5)** The model learns patterns in the data to create a functional model.

3. Model Evaluation & Validation: Test the trained model using Test Set **(1/5)** to assess its accuracy, precision, and performance in predicting outcomes

4. Deployment & Tuning: Deploy the trained Model, **Tune** with more Training/Testing phase when needed, **Retain** back to Data Collection phase if needed

Data Breadth – Collecting across a wide range or multiple contexts, Completeness for all scenarios often associated with quantitative data (Variety of Age Group)

Data Depth –Collecting comprehensive, nuanced, and detailed information about a smaller number of subjects or specific cases

Supervised Learning - The algorithm is given many examples of inputs and their correct outputs, learning relationship between them as the correct answer is given, used in Image Classifications

Unsupervised Learning - The algorithm works independently to find natural clusters, patterns, or associations within the data, used in Market analysis

Reinforcement Learning - The goal is to learn a sequence of actions that maximise the long-term reward, the "input" is continuous, real-time feedback that changes as the agent interacts with the world, used in Training autonomous vehicles, AI Gaming

Classification Model - Supervised machine learning where it tries to predict the discrete category or label of a given input data. (Logistic Regression, Decision Trees)

Regression Model - Supervised machine learning where it tries to predict a continuous numerical value. (Linear Regression, Ridge Regression)

Generative Model - Unsupervised machine learning that tries to create new data instances, generating New Images, Synthesized Text or Audio (GANs, VAEs, LLMs)

Overfitting - This is often because it learns aspects of the training set that are irrelevant to the task

Example: System that was trying to recognize but completely failed in the real world as all the pictures of the tanks are during cloudy day. The system learned the lighting conditions not the tanks themselves.

Minimise Overfitting - When it comes to doing machine learning, often the best thing to do is make sure the data set is good and when errors occur, that's when you start looking at the examples that aren't classified correctly and add new training examples or tell the model to ignore some features during training phase, but beware of **Underfitting**, it happens when it ignores too much, there is an unbalance

Biasness, Algorithm bias (Pick up extra detail from the training data which may reflect patterns which are not significant but yet considered significant),

Assumptions (Always revalidate the datasets or models created when there is clarity in the assumptions in future), **Over-reliance**, **Abuse**

Online Shopping: User HTTP Request triggers Server SQL Query to fetch product data, which is merged into HTML Template, sent to the Client via TCP/HTTPS, Client Add to Cart (State Change), Login (Security), Checkout via third party API, Card Provider confirms, Order stored in **Non-Volatile DB** for logistics, Delivery

Face Recognition: Camera captures RGB Pixels, ML algorithm performs Feature Extraction (Edges/Landmarks), Trained Classifier matches the pattern to a Name stored in a DB, Output Data via Cloud Sync (HTTPS)

Smart Watch: Accelerometer/Heart Rate sensors capture Analog signals, Analog to Digital Converter converts them, On-device ML classifies exercise type, Runs Formula to estimate calories, Update RAM before syncing to phone (Bluetooth) and Cloud DB via HTTPS.

Contact Tracing: Phone uses Bluetooth (PAN) to exchange Encrypted IDs, logs them in Local DB with timestamps, if positive test occurs, uploads the log to a Central Server via HTTPS for matching and sends back a notification for ID Match

Price Comparison: User search sends API/HTTP requests to external sites, which return JSON/XML data to be aggregated, sorted in a temporary DB, displayed via a Template for the user to select, after selection is done redirect to provider (HTTPS)

CPU – Central Processing Unit
RAM – Random Access Memory
DMA – Direct Memory Access
.exe – Executable files
.dll - Dynamic-Link Libraries
ISP – Internet Service Provider
IPv4/6 – Internet Protocol version 4/6
URL – Uniform Resource Locator
DNS – Domain Name Service
www. – World Wide Web (Web)
HTTP(S) – Hypertext Transfer Protocol (Secure)
HTML – Hypertext Markup Language
XML – Extensible Markup Language
JSON – JavaScript Object Notation
SMTP – Simple Mail Transfer Protocol
IMAP – Internet Message Access Protocol
POP3 – Post Office Protocol Version 3
API – Application Programming Interface
ASCII – American Standard Code for Info Interchange
DCT – Discreet Cosine Transform
RLE – Run Length Encoding
LZW – Lempel Ziff Welch
USB – Universal Serial Bus
Wi-Fi – Wireless Fidelity
IO – Input and Output
GUI – Graphical User Interface
DDoS – Distributed Denial of Service
STCF – Shortest Time to Completion First
PAN – Personal Area Network
LAN – Local Area Network
MAN – Metropolitan Area Network
WAN – Wide Area Network
TCP – Transmission Control Protocol
UDP – User Datagram Protocol
VoLP – Voice Over Internet Protocol
VoLTE – Voice Over Long-Term Evolution
SQL – Structured Query Language
GAN – Generative Adversarial Network
LLM – Large Language Model
VAE – Variational Autoencoder